Semiconductors 1

Classification of Materials :

- Conductors ($\alpha = +ve$, $T\uparrow \Rightarrow R\uparrow$)
- Insulators
- Semiconductors ($\alpha = -ve$, $T\uparrow \Rightarrow R\downarrow$)

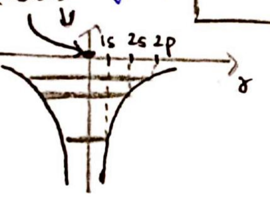
[Used in controlling flow of charges.]

Energy Band Diagram :

→ It is graphical representation of PE of e^- in positive field of nucleus.

1) For single isolated atom ⇒

PE of e^-, $\boxed{V = -\dfrac{K ze.e}{r}}$

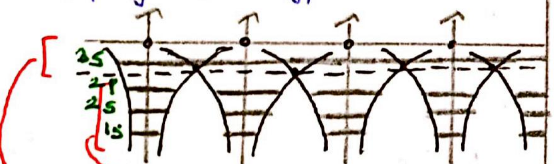


⊙ In a single atom, e^- revolves only in the influence of its own nucleus.

⊙ In atom, there are exactly defined energy levels of orbitals.

2) For Solids ⇒

→ In case of solids, due to atoms being closely packed, their orbitals overlap, resulting in formation of energy bands corresponding to an orbital, instead of exactly defined energy ~~bands~~ levels.



Conduction band (Contains free e^-)

Valence band (Contains immobile e^- that stay in their own atom).

Free electrons ⇒

e^- which exist under the influence of nucleus of other neigbouring atoms and can move throughout solid lattice.

Energy Bands :

→ In solids, due to overlapping, instead of discrete energy levels, energy bands exist.

→ Each energy band has its upper & lower energy value.

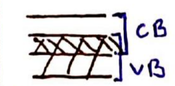
Fermi level ⇒

Uppermost energy level, upto which e^- are present in a solid intrinsically.

→ Whenever energy provided, fermi level e^- will excite first.

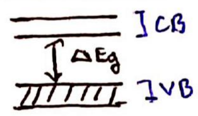
Band structures in Solids :

1) Conductors



(Overlapping bands)

2) Insulators



($\Delta E_g > 5eV$)

3) Semiconductors

($\Delta E_g < 2.2eV$)

ΔE_g → Fermi energy gap / Forbidden Energy Gap

↳ Thus, they are able to conduct current at room temp., & their conductivity ↑ses on ↑sing temp.

Types of Semiconductors :

1) Intrinsic S.C ⇒

Pure IV Grp elements ⇒ Ge & Si

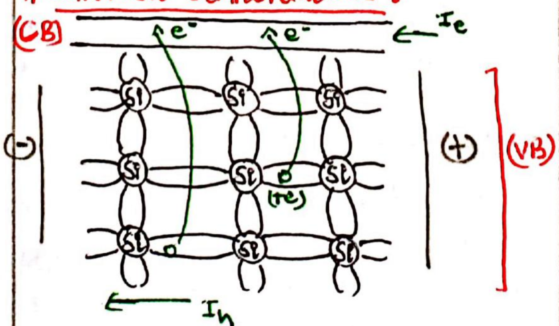
($\Delta E_g = 0.7eV$) ($\Delta E_g = 1.1eV$)

2) Extrinsic SC ⇒

Pure SC doped with impurities of Grp III or Grp V elements

Grp III ↓ (p-type SC) Grp V ↓ (n-type SC)

Intrinsic Semiconductors :



→ A hole has +e charge.

→ Approx, 10^8 Si atoms will contribute one electron-hole pair.

Total current ⇒

$$\boxed{I = I_e + I_h}$$

Concentration of intrinsic charge carrier ⇒

$$\boxed{n_i = A T^3 e^{-\Delta E_g / KT}}$$

$$\boxed{n_i = n_0 e^{-\Delta E_g / KT}}$$

(n_0 is a temp. dependent constant)

Conductivity ⇒

We can write, $\boxed{\sigma = n_i e \mu}$

where,

$$\sigma_T = \sigma_e + \sigma_h$$

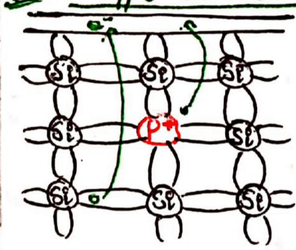
$$\boxed{\sigma_T = n_e \cdot e \cdot \mu_e + n_h \cdot e \cdot \mu_h}$$

Also, in intrinsic SC

$$\boxed{n_e = n_h = n_i}$$

Extrinsic Semiconductors :

(I) n-type semiconductors ⇒



⊙ Grp V elements doped as impurities

⊙ Electrons are majority charge carriers.

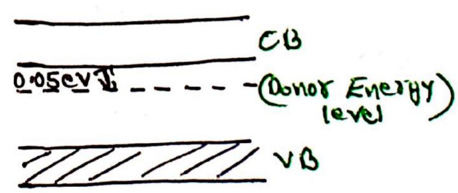
⊙ Holes are minority charge carriers

$$\boxed{n_e >>> n_h}$$

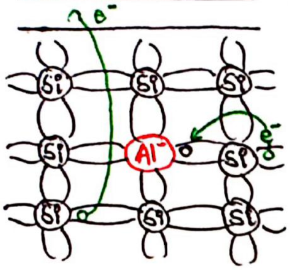
$$\Rightarrow \boxed{I \simeq I_e} \quad (I_h \to 0)$$

⊙ 'P' exist as +ve ion in lattice due to loosing extra e^- at Room temp.

⊙ The extra e^- lie in donor energy level.



(II) p-type Semiconductors ⇒

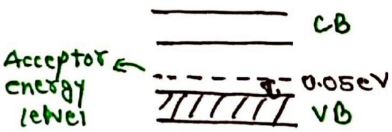


- Grp III elements doped as impurity.
- Holes are majority charge carriers
- Electrons are minority charge carriers.

$$\boxed{n_h >>> n_e}$$

$$\Rightarrow \boxed{I \simeq I_h} \quad (I_e \to 0)$$

- e^- vacancy of Al is filled by neighbouring e^-, & 'Al' exist as −ve ion in lattice.
- Before C.B, electrons fill the acceptor energy level.



(*) Carrier Concentration ⇒

by law of mass action

p type ⇒ $n_h >>> n_e$
n-type ⇒ $n_e >>> n_h$
$\Big\} \rightarrow \boxed{n_e \cdot n_h = n_i^2}$ ***

→ For p-type SC, if acceptor impurity conc. is N_A

$$\Rightarrow \boxed{n_h \simeq N_A}$$

$$\boxed{n_e = \frac{n_i^2}{N_A}}$$

→ For n-type SC, if donor impurity conc. is N_D

$$\Rightarrow \boxed{n_e \simeq N_D}$$

$$\boxed{n_h = \frac{n_i^2}{N_D}}$$

(*) Electrical Conduction :→

In metals ⇒

$$J = ne v_d \qquad (\sigma = ne\mu)$$

$$J = ne\mu E \Rightarrow \boxed{J = \sigma E}$$

In Semiconductors ⇒

$$J = (\sigma_e + \sigma_h) E$$

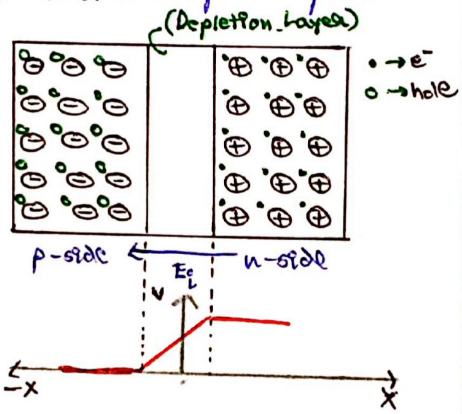
$$J = (n_e \mu_e e + n_h \mu_h e) E$$

$$\boxed{J = eE(n_h \mu_h + n_e \mu_e)}$$

↳ For both extrinsic & intrinsic semiconductors.

p-n junction formation :

→ Formed by doping equal half of pure SC with donor & acceptor impurities.



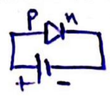
→ Due to +ve & −ve ions, an internal electric field exist which holds e & h in their respective regions & forms a 'Depletion Layer' at junction.

→ Depletion layer is free from any type of charge carrier.

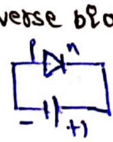
ckt Symbol & Biasing :

→ Study of application of potential diff across diode is biasing.

↳ Forward bias (Diode behaves as closed circuit)



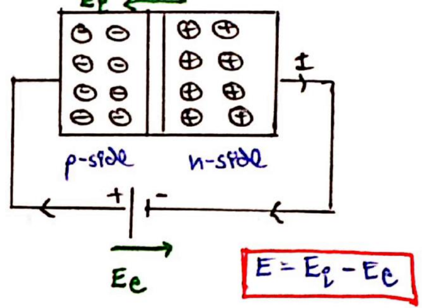
↳ Reverse bias (Diode behaves as open ckt)



(I) Forward Biasing :→

→ Width of depletion layer decreases.

→ Ext. battery ↓ses potential diff. of depletion layer in opp dirⁿ & current starts flowing.

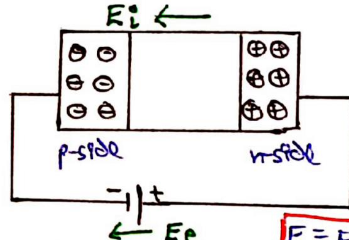


$$\boxed{E = E_i - E_e}$$

(II) Backward Biasing :→

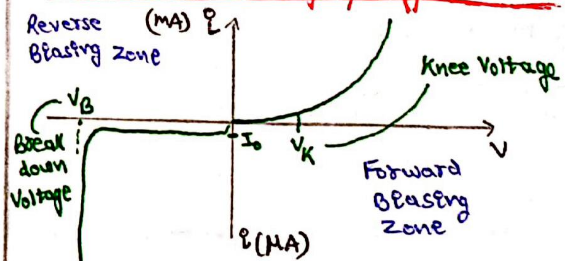
→ Width of depletion layer increases.

→ Ext. battery ↑ses the barrier potential due to same dirⁿ of EF and no current passes (almost).



$$\boxed{E = E_i + E_e}$$

V-I characteristic of p-n junc. Diode :



- During forward biasing, V-I is non-linear violing ohm's law.
  ↳ p-n junc. diode is non-ohmic device.
- Current in FB is given by,

$$\boxed{i = I_0 (e^{eV/KT} - 1)}$$

(I_0 = Reverse saturation current)

For high voltages $e^{eV/KT} >>> 1$

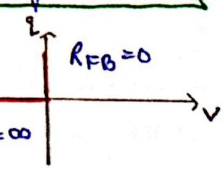
$$\Rightarrow \boxed{i = I_0 \cdot e^{eV/KT}}$$

- Dynamic Resistance

$$\boxed{R_D = \frac{\Delta V}{\Delta I} = \frac{1}{\text{Slope of VI curve}}}$$

- For ideal diode ⇒

RB → Open ckt
FB → Closed ckt



Breakdown of Diode :

→ In reverse biased p-n junc. diode, after a certain high voltage (-ve), the current starts flowing exponentially.

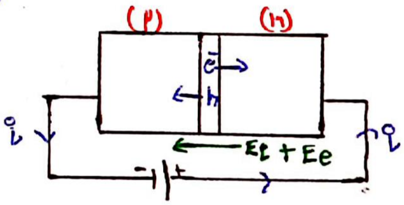
→ This phenomenon is 'Breakdown'

Takes place by following two mechanisms.

(I) Zener Breakdown :→

→ It happens when depletion layer is very thin & diode is heavily doped.

→ In depletion layer, bond breaking starts due to very high EF & excess electron-hole pairs are generated.

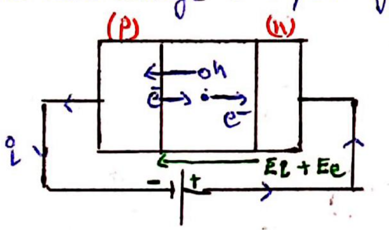


(II) Avalanche Breakdown :→

→ It takes place when depletion layer is thick & SC is moderately doped.

→ The minority e⁻ of p-side gets accelerated due to high EF, & in the process hits other bounded e⁻ of depletion layer.

→ This causes huge e-h pair generation.

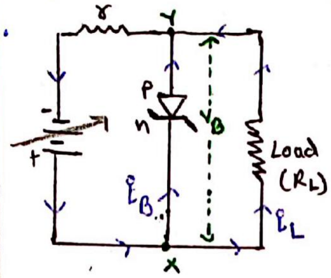


Zener Diode :

↳ In FB, it behaves like a normal diode.

↳ In RB, at breakdown, it behaves like a voltage stabiliser.

→ In normal reverse biasing, it acts as open ckt. But after breakdown, it allows excess current to pass through it while maintaing the same voltage V_B & prevents load device getting damaged.



V_B → Zener voltage / Breakdown Voltage

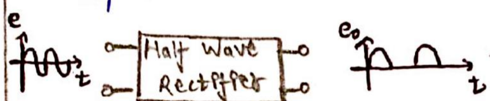
Rectifiers :

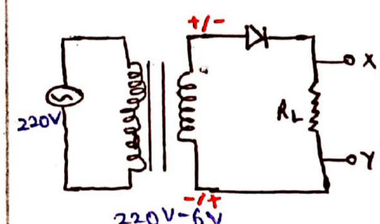
↳ Converts AC into unidirectional pulsating DC.

(I) Half Wave Rectifier :→

→ uses only one diode.

→ Gives only half of the original as output.





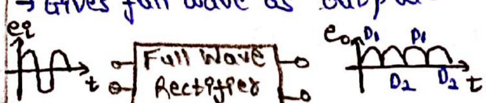
Positive Half cycle → FB → Current conducted

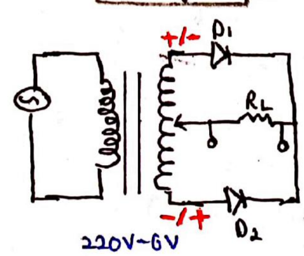
Negative Half cycle → RB → Current not conducted

(II) Full Wave Rectifier :→

→ Use two diodes & a centre tap (which makes it very expensive)

→ Gives full wave as output.



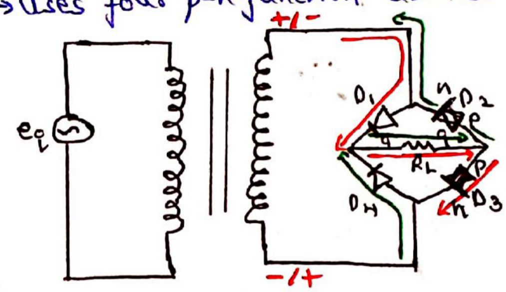


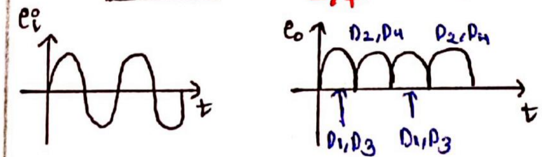
→ Alternatively, either of one diod is forward biased at atime & conducts current.

(III) Bridge Rectifier :→

→ Cheaper alternative of full wave rectifier.

→ Uses four p-n junction diodes.





Currents & efficiency of Rectifiers :→

	Half Wave	Full Wave
Avg. Current	$\frac{I_{max}}{\pi}$	$\frac{2I_{max}}{\pi}$
rms Current	$\frac{I_{max}}{2}$	$\frac{I_{max}}{\sqrt{2}}$
Efficiency	$\frac{0.406}{(1+r_d/R_L)}$ ⇓ 40.6%	$\frac{0.812}{(1+r_d/R_L)}$ ⇓ 81.2%

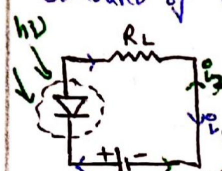
$$\eta = \frac{(I_{avg}/I_{rms})^2}{(1+r_d/R_L)}$$

For ideal diode ($r_d \to 0$)

Photo Diode :

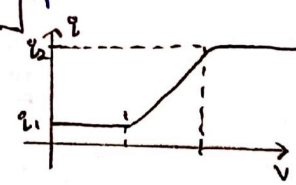
→ Used in reversed bias.

→ Light incident on depletion layer, which breaks covalent bonds & flows high amount of current.



Initially current → i_1

After incident current → i_2

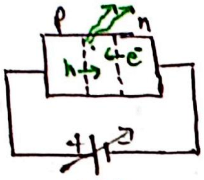


Light Emitting Diode (LED) :

→ It is a diod in forward bias.

→ When voltage greater than knee voltage (V_K) is applied, then electrons drop from CB in VB to combine with holes to use depletion layer.

→ This transition of e^- from CB to VB emits visible radiation.



$V > V_K$

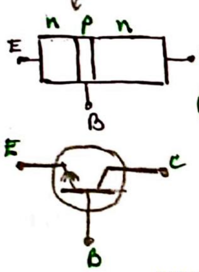
For Si, $\Delta E_g = 1.14\,eV$

$$\boxed{\lambda_{emit} = \frac{12431}{1.14} \simeq 10800\,\mathring{A}}$$

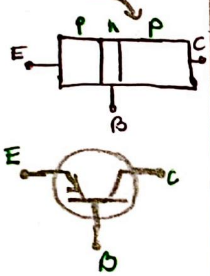
Transistors :

↳ 3 terminal, 2 junction device

↳ Generally, we discuss two types of transistors.



(Arrow is always towards n-side)

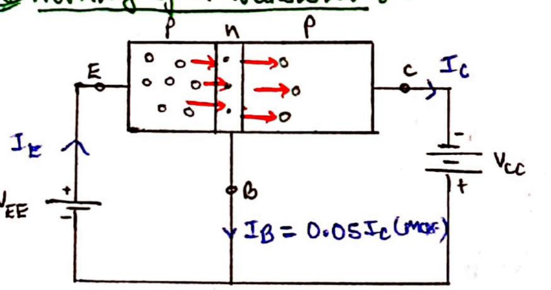


Emitter ⇒ ~~Moderately~~ Highly doped, medium in size

Base ⇒ Very less doped, thinnest region

Collector ⇒ Moderately doped, thickest region

Working of a Transistor :→



→ Generally, Base-emitter ⇒ ~~Reverse~~ Forward biased

Base-collector ⇒ ~~Forward~~ Reverse biased

→ Due to FB, holes from emitter are accelerated towards base, where they combine with very few e^- (as low doping)

→ Rest of holes get flowed to collector.

→ Due to RB, the holes of collector are already far from depletion layer.

→ So, holes coming from base gets more time to accelerate & flow in collector.

→ Almost 95%–98% holes pass through base, w/o getting recombined with e^-

So,

$$\boxed{I_E = I_B + I_C}$$

$$\Rightarrow \boxed{I_E \simeq I_C} \quad (I_B \lll I_C)$$

DC gains of a Transistor :→

(i) Base current amplification factor

$$\boxed{\beta = I_C / I_B}$$

(ii) Emitter current amplification factor

$$\boxed{\alpha = I_C / I_E}$$

We know, $I_E = I_C + I_B$

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$$

$$\Rightarrow \boxed{\alpha = \frac{\beta}{\beta+1}} \quad \& \quad \boxed{\beta = \frac{\alpha}{1-\alpha}}$$

Transistor as Amplifier :

→ Generally, we discuss only 'Common Emitter Amplifier'

Input ⟶ B & E

Output ⟶ E & C

Operating regions of a Transistor ⇒

(i) Active Region : BE junc. → FB, BC junc. → RB

(ii) Cut-off Region : BE & BC junc. in Reverse bias

(iii) Saturation Region : BE & BC junc. in forward bias

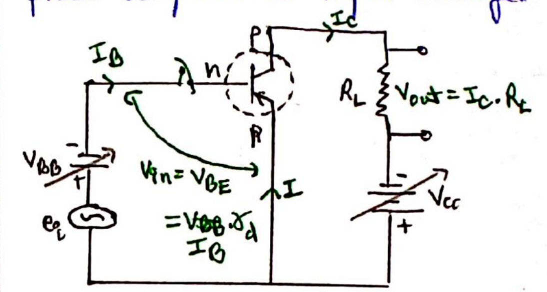
Working :→

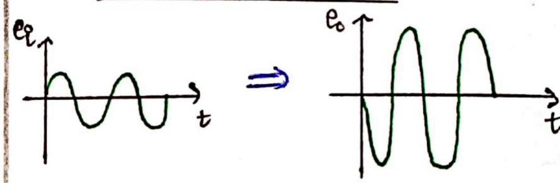
→ Biasing of base effect emitter to collector current.

More FB ⇒ $I_B \uparrow$ ⇒ $I_C \uparrow$

→ Also, I_C fluctuates same as I_B, just its magnitude ↑ses.

→ And, output voltage is 180° out of phase compared to input voltage.





Gains in Common Emitter Amplifier :→

1) AC Current Gains ⇒

$$\boxed{\beta_{AC} = \frac{\Delta I_C}{\Delta I_B}}$$

2) AC Voltage Gains ⇒

$$\boxed{A_V = \frac{\Delta V_o}{\Delta V_{in}} = \frac{\Delta I_C \cdot R_L}{\Delta I_B \cdot r_{BE}} = \beta_{AC} \cdot \frac{R_L}{r_{BE}}}$$

[r_{BE} = Dynamic resistance across Base emitter junc.]

3) AC Power Gain ⇒

$$\boxed{A_P = \frac{\Delta I_C^2 R_L}{\Delta I_B^2 \cdot r_{BE}} = \beta_{AC}^2 \cdot \frac{R_L}{r_{BE}}}$$

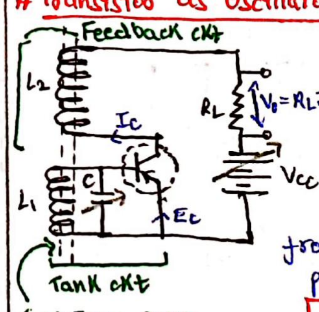
4) Transconductance ⇒

$$\boxed{g_m = \frac{\Delta I_C}{\Delta V_{BE}}} = \frac{\Delta I_C}{\Delta I_B} \times \frac{\Delta I_B}{\Delta V_{BE}}$$

$$\boxed{g_m = \beta_{AC} \cdot \frac{\Delta I_B}{\Delta V_{BE}}}$$

Transistor as Oscillator :

[Electrical device producing alternating signal from DC input]



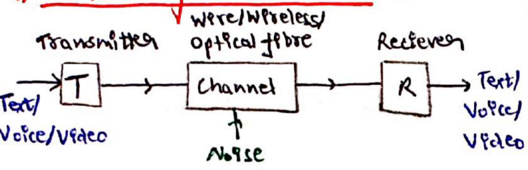
⊙ Tank ckt has inductor & cap. in parallel to transistor that draw energy from V_{CC} & generate pulsating LC-voltage.

$$\boxed{f = \frac{1}{2\pi\sqrt{LC}}}$$

⊙ Feedback ckt has another inductor which ~~show~~ reverse phase of signal produced by L_1 and maintain magnetic energy in L_1 const.

Communication System

Elements of Communication :



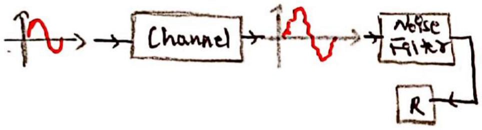
(*) Transducer :→

→ Present in transmitter as well Reciever.

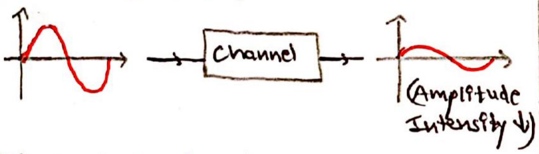
→ Converts info. into electrical signal & vice-versa.

→ Eg: Mic, fax, speaker, Monitor, etc.

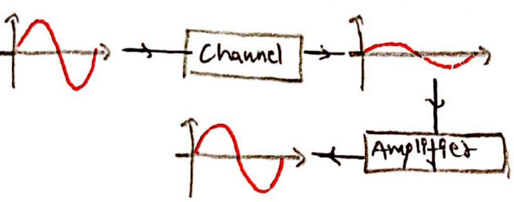
(*) Noise :→ unwanted signals disturbing transmission process.



(*) Attenuation :→ Loss of signal strength during propogation in channel.

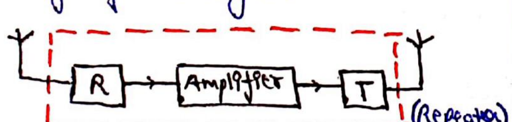


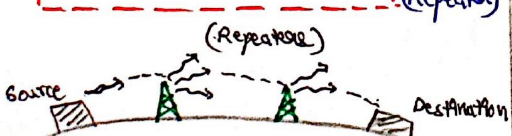
(*) Amplification :→ Increase in amplitude of a signal.



(*) Range :→ Largest distance over which signal can be transmitted from source to destination with sufficient signal strength.

(*) Repeater :→ Devices used to increase range of the signal.





Bandwidth of Signals & Transmission medium :

(*) Range of freq. of info. signals ⇒

		Bandwidth
Speech signal	300Hz - 3100Hz	2800Hz
Music signal	20Hz - 20KHz	20KHz (Almost)
Video signal		4.2 MHz
T.V. signal		6 MHz

(*) Operating frequency of channels ⇒

Wire → Coaxiable cable, 10^9 to 10^{10} Hz

Free space → 10^8 Hz to 10^{10} Hz

Optical Fibre → 10^{12} Hz to 10^{15} Hz

Imp

$$\boxed{\text{No. of channels} = \frac{\text{Total Bandwidth of channel}}{\text{Bandwidth per channel}}}$$

Transmission of EM waves :

$$\boxed{P_{rad.} \propto \left(\frac{l}{\lambda}\right)^2}$$

(l → Length of antena)

→ For optimum power transmission of signals,

$$\boxed{\text{size of antena}(l) \geq \frac{\lambda}{4}}$$

→ High frequency of signals required for transmission.

$$\boxed{f\uparrow \Rightarrow \lambda\downarrow}$$

Practical Antena size

$\lambda\downarrow \Rightarrow P_{rad.}\uparrow$

Propogation Modes of EM waves :

1) Ground Waves :→

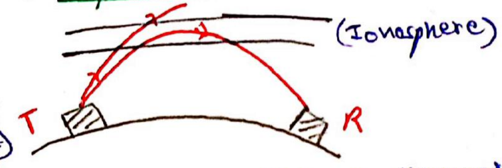
(surface wave)

→ Mode of propogation is called surface wave.

→ Waves are attenuated due to absorption by earth surface.

→ used for low range transmissions

2) Sky wave :→



→ It is a long distance propogation done by Ionospheric Reflection.

→ Ionosphere : 65Km to 400Km

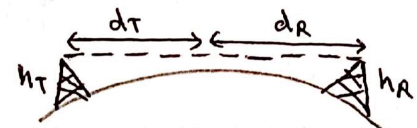
→ 3 to 30MHz ⇒ Ionosphere reflects

f > 30MHz ⇒ Penetrates Ionosphere

3) Space Wave :→

→ For (f > 40MHz), line of sight propogation is done.

→ L.O.S must be clear for transmission and reciever antenna.



Radi of earth = R_0

And, $R_0 \ggg h_T$ & $R_0 \ggg h_R$

Now,

$$d_T = \sqrt{(h_T + R_0)^2 - R_0^2}$$

$$d_T = \sqrt{h_T^2 + R_0^2 - R_0^2 + 2h_T R_0}$$

$$\Rightarrow \boxed{d_T \simeq \sqrt{2h_T R_0}} \quad \& \quad \boxed{d_R \simeq \sqrt{2h_R \cdot R_0}}$$

→ Max. L.O.S distance

$$\boxed{\begin{aligned} d &= d_T + d_R \\ d &= \sqrt{2h_T R_0} + \sqrt{2h_R \cdot R_0} \end{aligned}}$$

P.T.O

Problems in transmission of baseband signals:

$l \geq \frac{\lambda}{4}$

Antenna size
- → At low frequency, λ is too high
- → Impractical Antenna size.

Radiation Power

$P_{rad.} \propto \frac{1}{\lambda^2}$

Low $f \Rightarrow \lambda \uparrow$

$\Rightarrow P_{rad.} \downarrow$

Mixing of signals

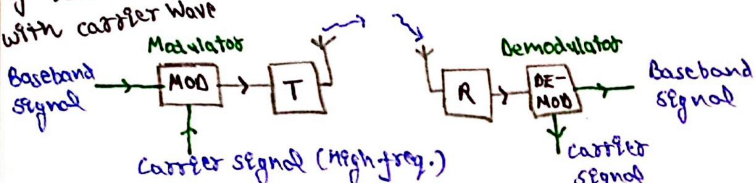
If several voice channels are broadcasted at same time, all signals gets mixed up.

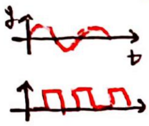
Solution: Using communication at high frequency and allotting a band of frequencies to each message signal for its transmission.

Modulation of baseband signals & carrier waves:

(Modulating signal) (High frequency signals)

Superposition of baseband with carrier wave



→ sinusoidal  $\boxed{y_c = A_c \sin(\omega_c t + \phi_c)}$

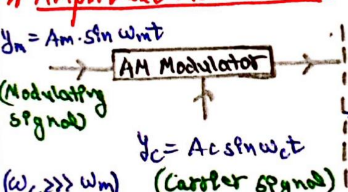
→ Pulsating

Types of Modulation:

$y_c = A_c \sin(\omega_c t + \phi_c)$ → During Modulation: A_c, ω_c or ϕ_c can be controlled by modulating signal or info. signal

1) Amplitude Modulation — Amp. of carrier wave is varied according to modulating signal.
2) Frequency Modulation
3) Phase Modulation

Amplitude Modulation:



Amp. varies as per modulating signal

$y = \underbrace{(A_c + A_m \sin\omega_m t)} \sin\omega_c t$

$\Rightarrow y = A_c\left(1 + \frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t$

↳ $\boxed{\text{Modulation index} = \frac{A_m}{A_c} = \mu}$

$$\Rightarrow \boxed{y = A_c \sin\omega_c t + \frac{\mu A_c}{2}\cos(\omega_c - \omega_m)t - \frac{\mu A_c}{2}\cos(\omega_c + \omega_m)t}$$

Carrier frequency: ω_c

Lower side frequency: $\omega_c - \omega_m$

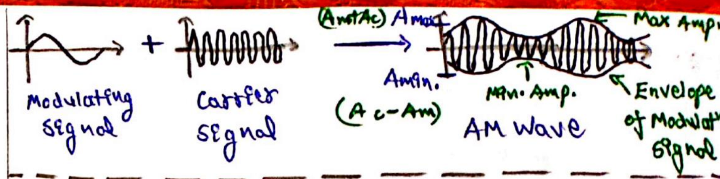
Upper side frequency: $\omega_c + \omega_m$

$\boxed{\text{Bandwidth} = 2\omega_m = 2f_m}$

Note:

Modulation index,

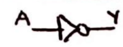
$\mu = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$



Logic Gates:

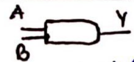
(I) Basic Gates:

1) NOT Gate ⇒



A	$Y = \bar{A}$
1	0
0	1

2) AND Gate



A	B	$Y = A \cdot B$
1	0	0
1	1	1
0	1	0
0	0	0

3) OR Gate

A	B	$Y = A + B$
1	0	1
1	1	1
0	1	1
0	0	0

(II) Rules of Boolean Algebra:

1) $A + 0 = A$
2) $A + A = A$
3) $A \cdot A = A$
4) $A \cdot \bar{A} = 0$
5) $\overline{(\bar{A})} = A$

(III) Demorgan's Theorem:

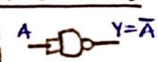
$\boxed{\overline{A \cdot B} = \bar{A} + \bar{B}}$ $\boxed{\overline{A + B} = \bar{A} \cdot \bar{B}}$

(IV) Universal Gates:

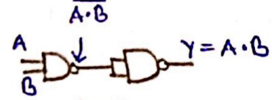
1) NAND Gate (AND + NOT) ⇒

$(Y = \overline{A \cdot B})$

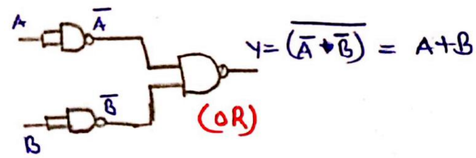
A	B	$\overline{A \cdot B}$
0	1	1
0	0	1
1	0	1
1	1	0



(NOT)



(AND)



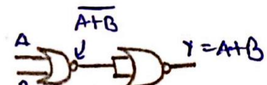
(OR)

2) NOR Gate (NOT + OR) ⇒

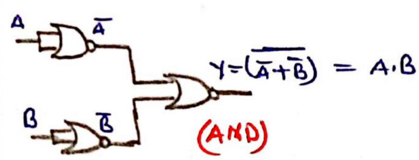
$Y = \overline{A + B}$

A	B	$\overline{A + B}$
0	1	0
0	0	1
1	1	0
1	0	0

(NOT)

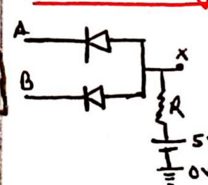


(OR)



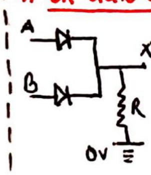
(AND)

AND Gate using Diode:



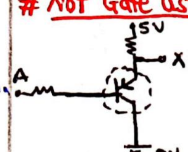
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate using Diode:



A	B	X
0	0	0
1	0	1
0	1	1
1	1	1

NOT Gate using Transistor:



A	X
0	1
1	0